Exploring the polarimetric signatures of a potential binary supermassive black hole system in Mrk 231

Presented by

BIEDERMANN Julie

MARIN Frédéric
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Observatoire Astronomique de Strasbourg

22-23 January 2025

Workshop HWO/Pollux - 22-23 January 2025 - Meudon

I. Introduction - Markarian 231 (Mrk 231)

Mrk 231

- **Ultra luminous infrared galaxy**
  - z = 0.042 (~ 175 Mpc)

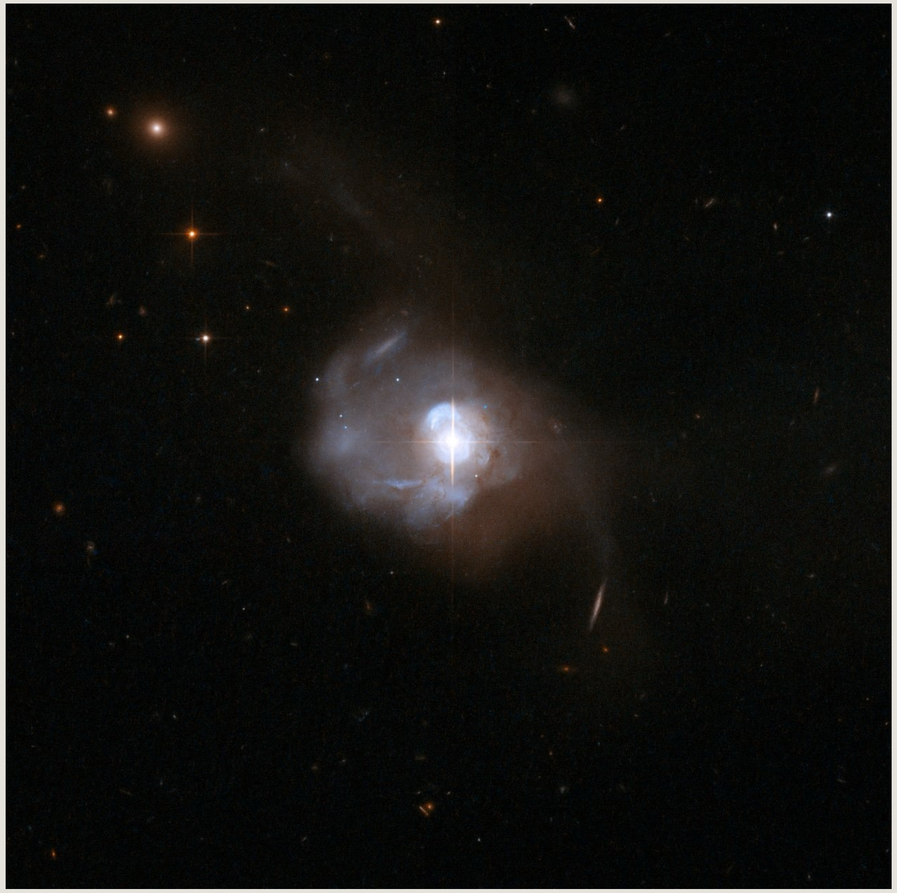
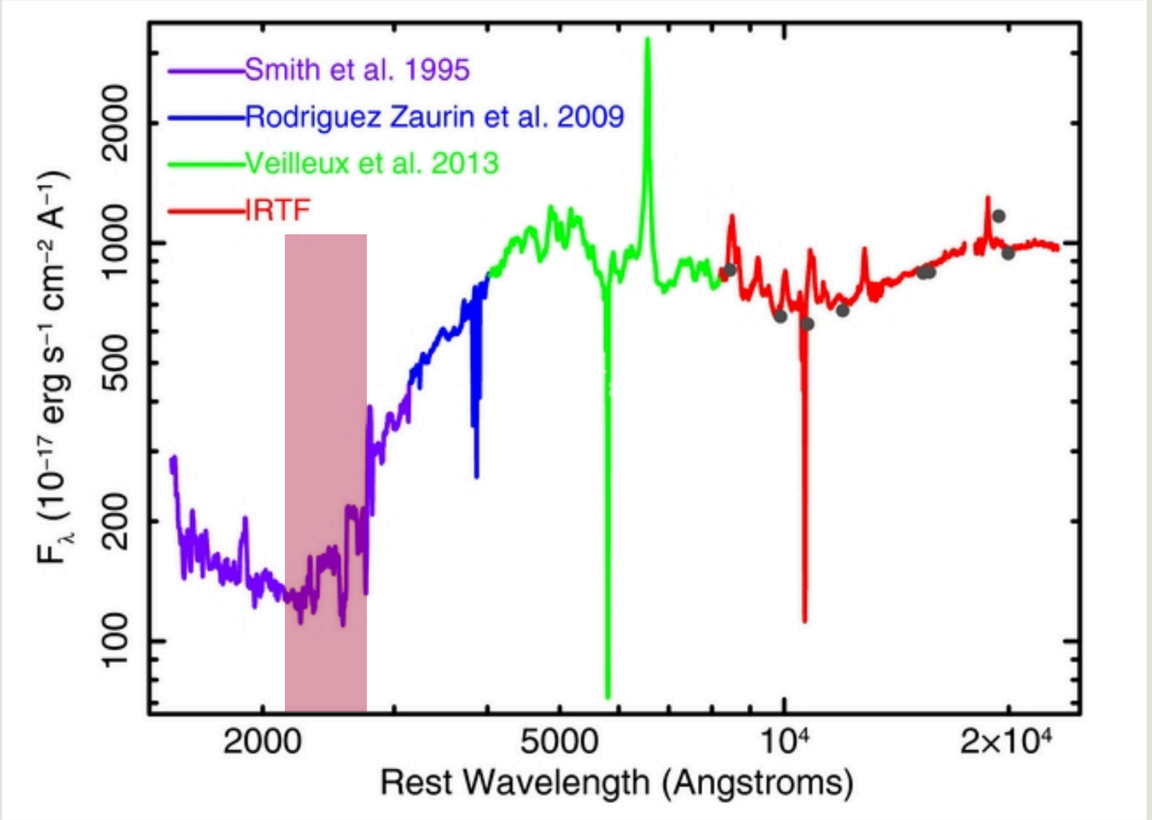


Image of Mrk 231, *Hubble Space Telescope*

I. Introduction - Markarian 231 (Mrk 231)

Mrk 231 - unique spectrum

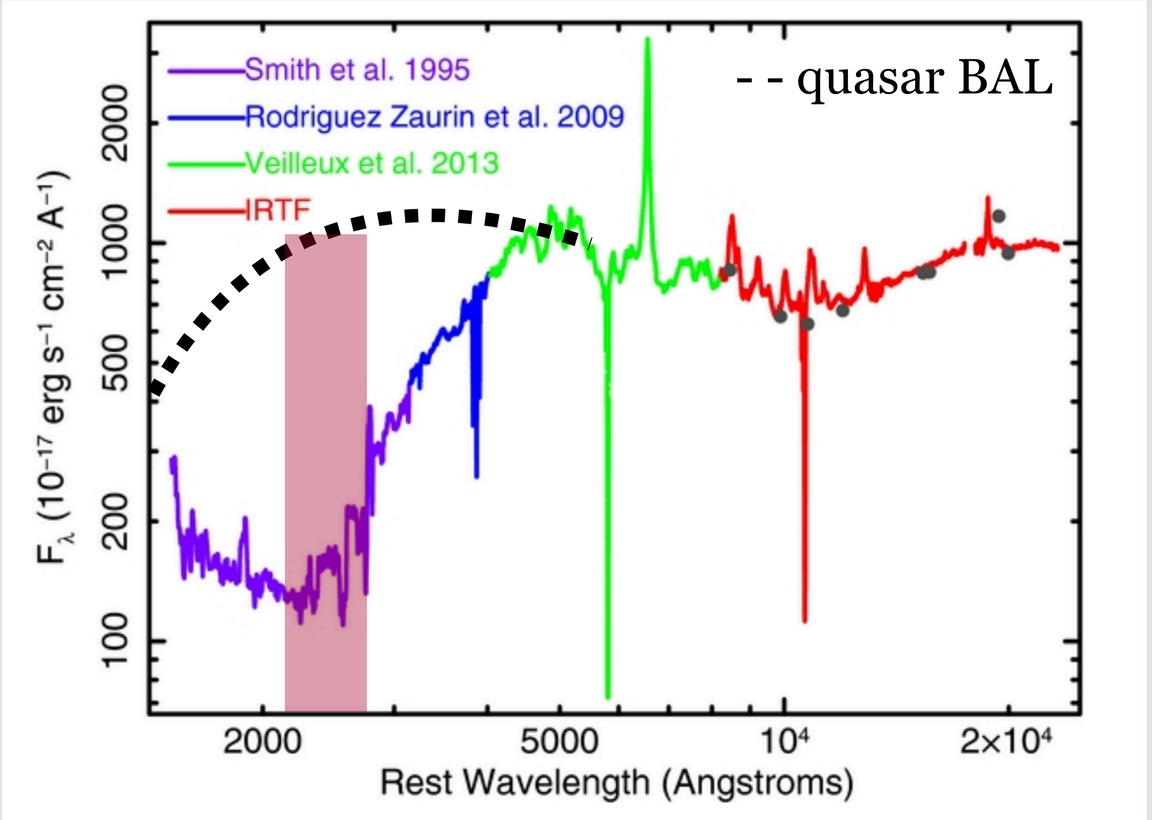
- **Ultra luminous infrared galaxy**
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- **Unique UV- optical spectrum**
  - Cut off ~ 2400 Å



Mrk 231 UV-IR spectrum, *Leighly et al. (2017)*

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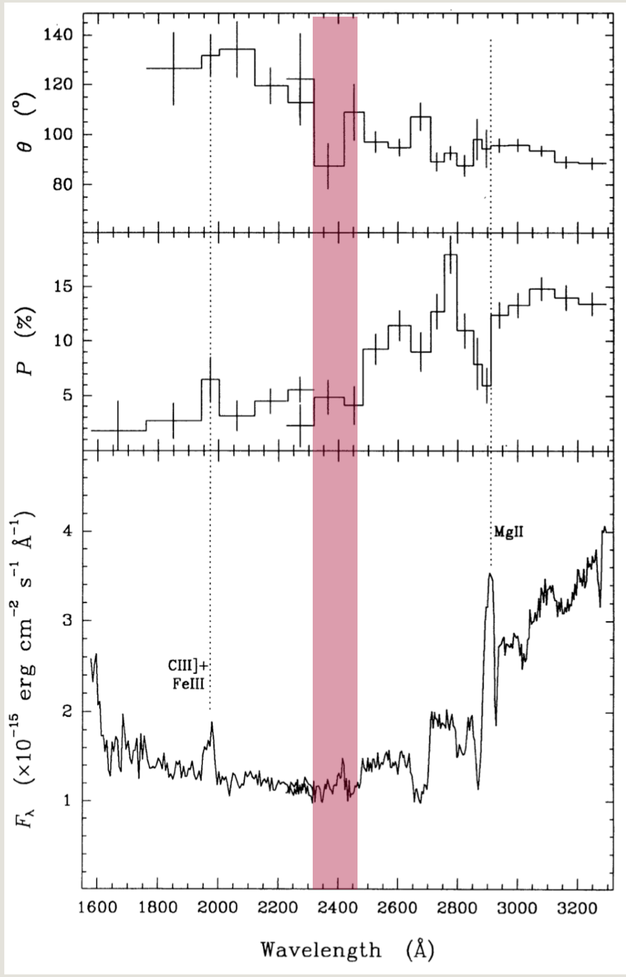


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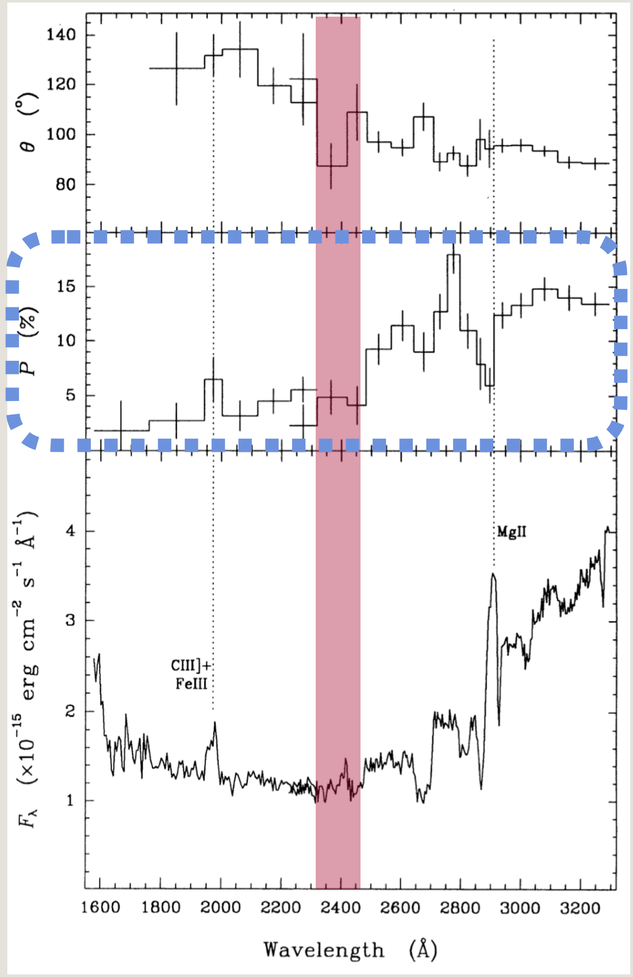


Spectropolarimetric data of Mrk 231, HST/FOS, *Smith et al. (1995)*

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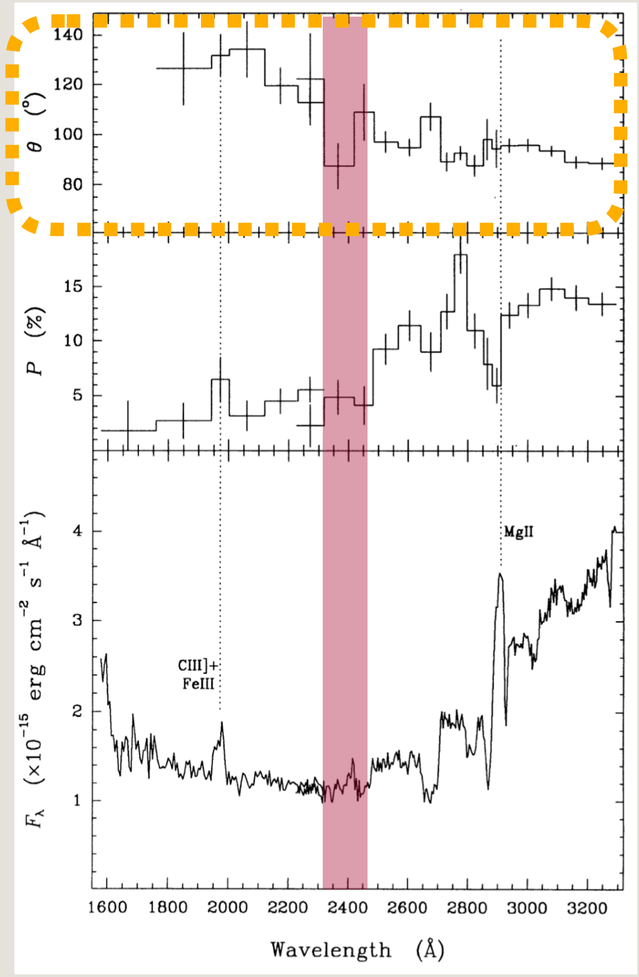


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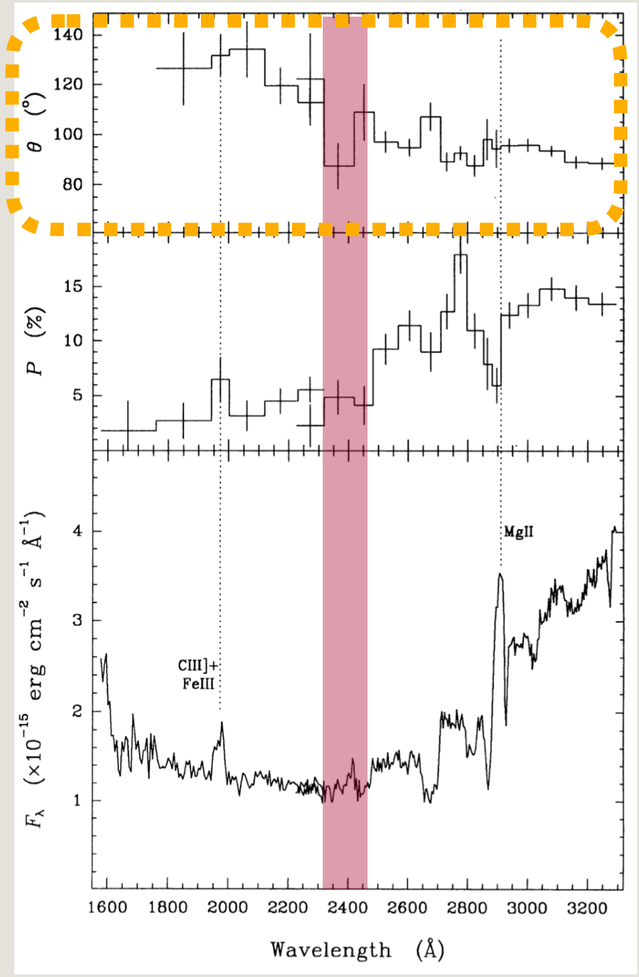
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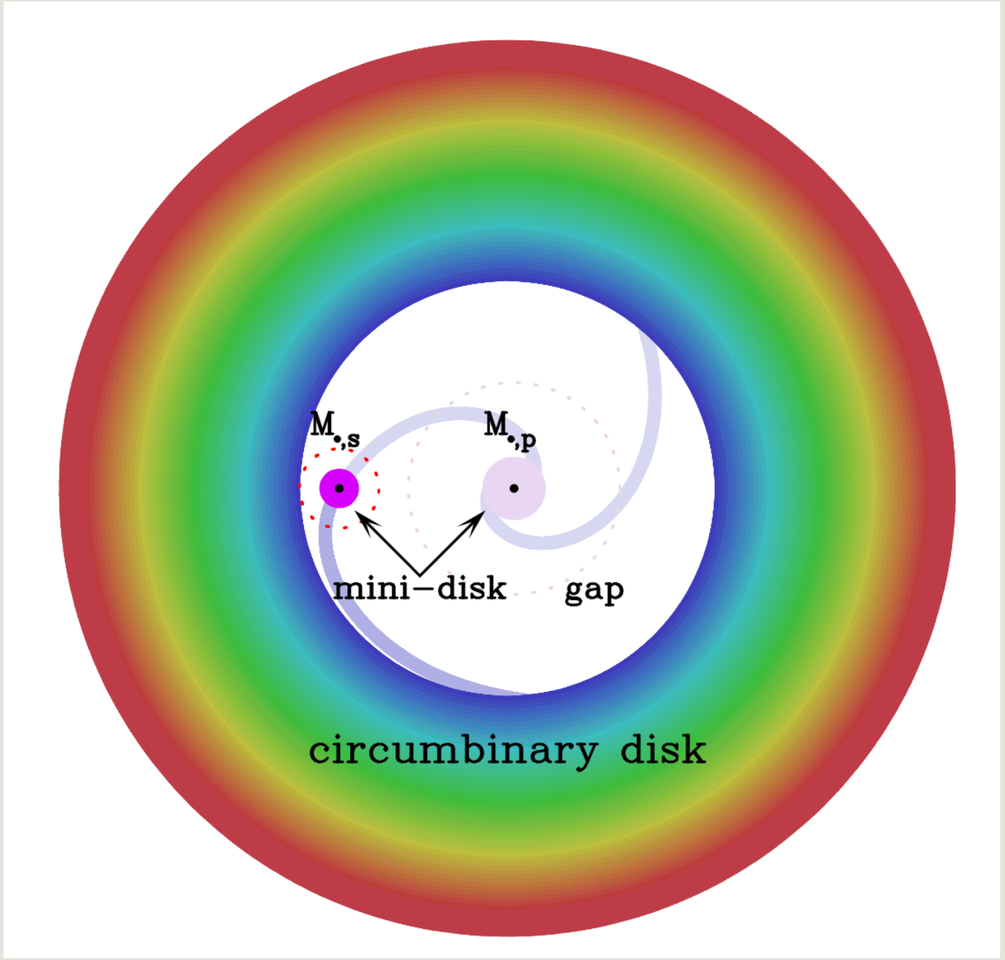
➡ **Two sources ?**



Spectropolarimetric data of Mrk 231, HST/FOS, *Smith et al. (1995)*

Mrk 231 - Hypothesis for a supermassive black holes binary (SMBH) - Yan and al (2015)

- **Binary SMBH in the core of Mrk 231**
  - To explain the cut off in the UV spectrum

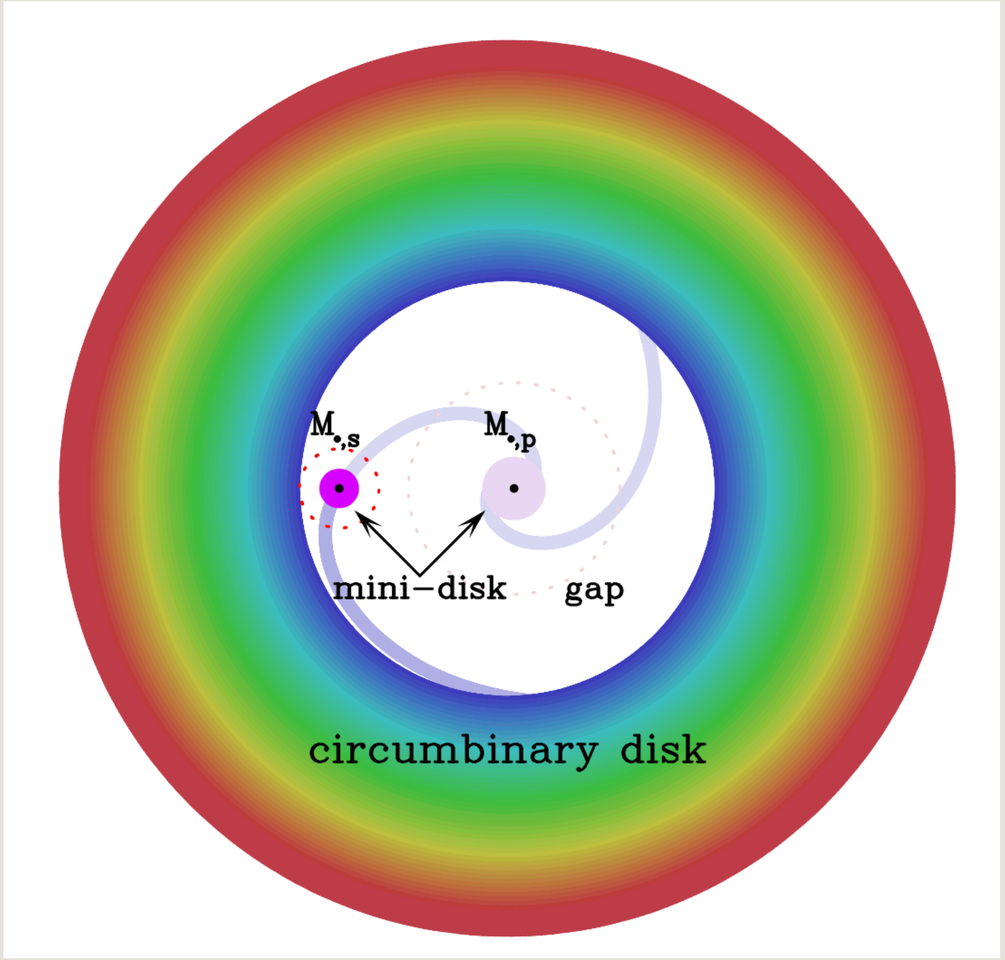


Accretion disk diagram of a supermassive black hole binary system, *Yan et al. (2015)*

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- **Binary SMBH in the core of Mrk 231**
  - To explain the cut off in the UV spectrum
- **GOAL**
  - Reproduce the Spectral Energy Distribution (SED)
  - Analyse the response of the model in polarization
    - ➡ **Never studied !**



Accretion disk diagram of a supermassive black hole binary system, *Yan et al. (2015)*

II . Modele Spectral Energy Distribution - SED Accretion disk

Unique supermassive black hole SED

- Geometrically thin and optically thick disk model

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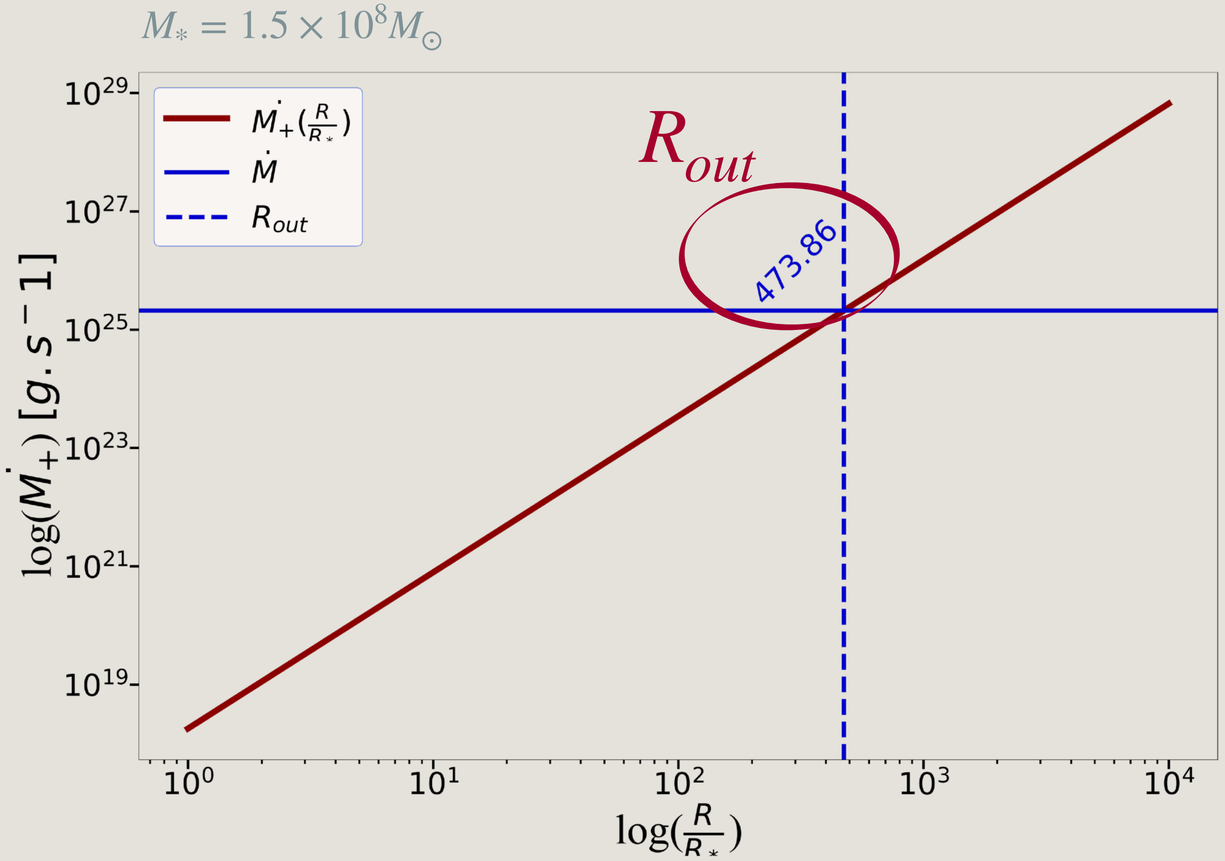
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- Theoretical radius - Critical accretion rate $\dot{M}$
  - Disk becomes unstable : $\dot{M} > \dot{M}_{+}$ *Lacosta (2016)*



LOG-LOG critical accretion rate as a function of the normalized radius
Biedermann et al. (submitted)

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- Parameters for the black hole model

Viscosity

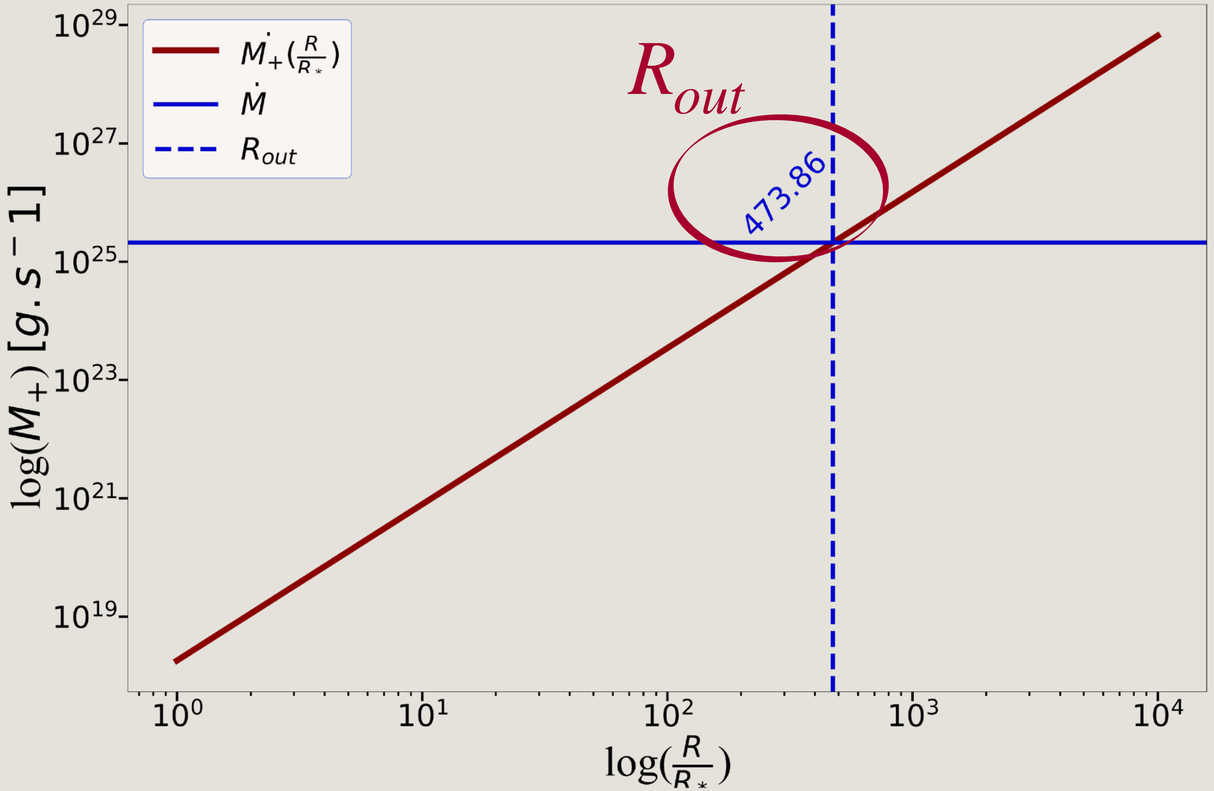
Inclination angle

$$\boldsymbol{\theta} \equiv \{M_*, \alpha, p, \theta\}$$

Black hole mass

Accretion parameter

$M_* = 1.5 \times 10^8 M_\odot$



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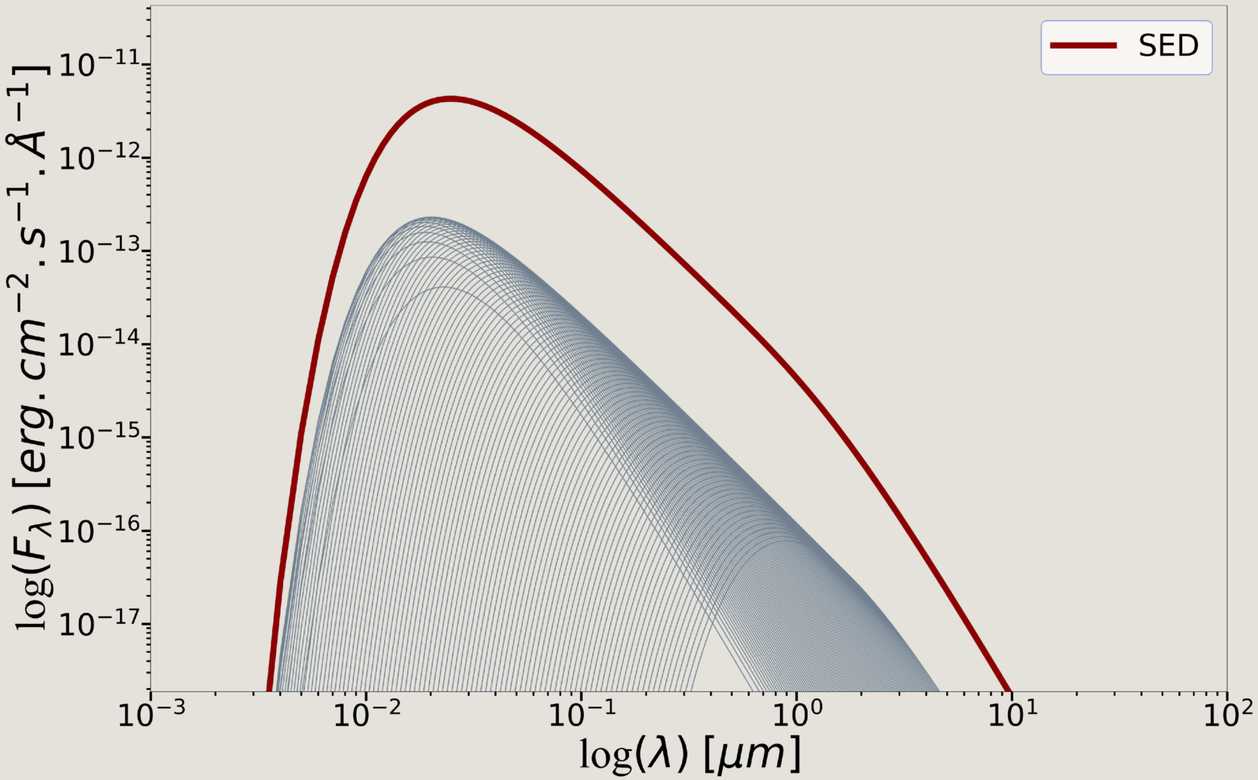
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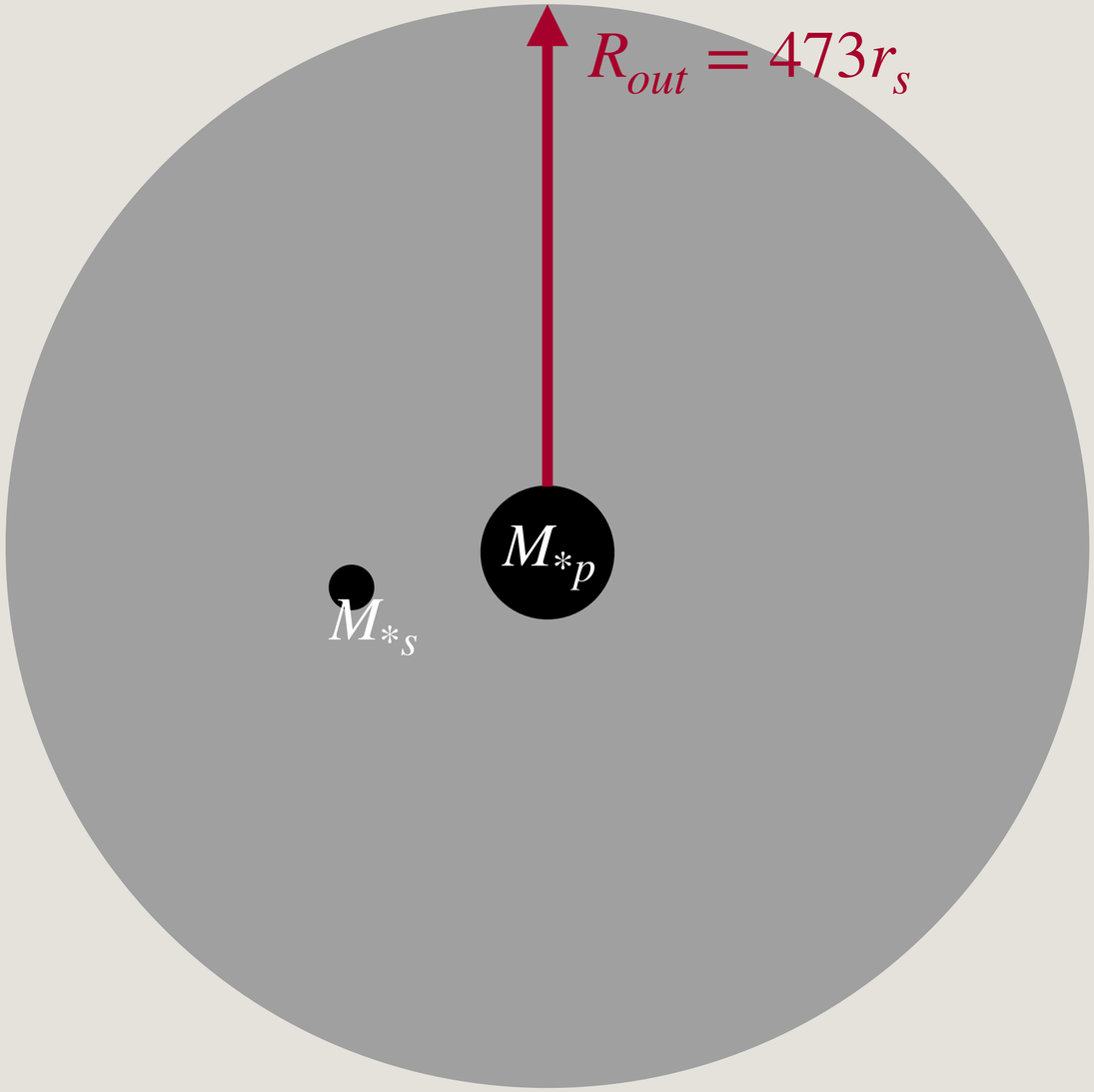


LOG-LOG Flux a function of the wavelength $M_* = 1.5 \times 10^8 M_\odot$

Biedermann et al. (submitted)

Binary supermassive black hole SED

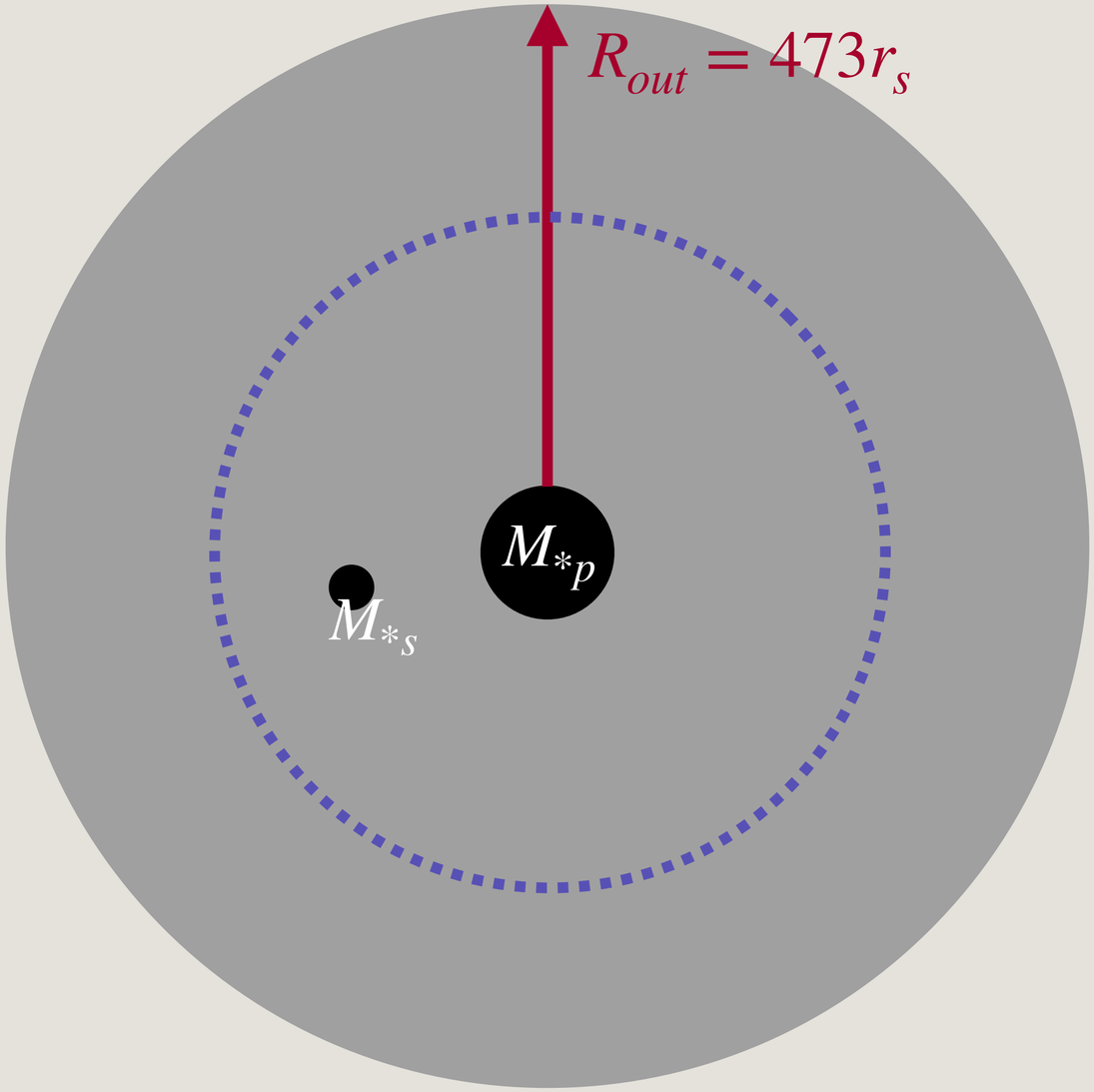
- Let's add a secondary SMBH



II . Modele Spectral Energy Distribution - Accretion disk

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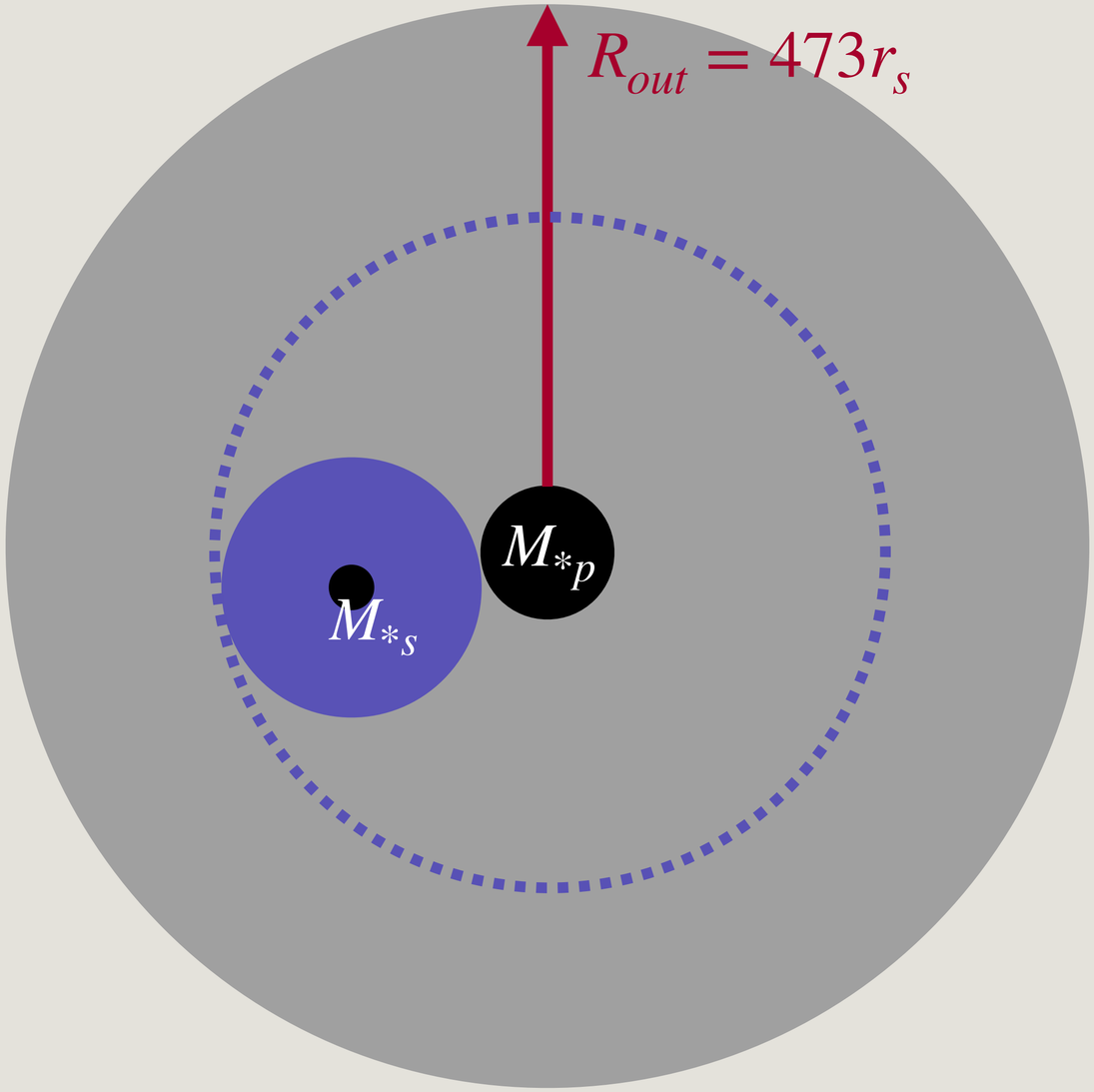
- Let's add a secondary SMBH
  - Inner radius of the primary SMBH



II . Modele Spectral Energy Distribution - Accretion disk

Binary supermassive black hole SED

- Let's add a secondary SMBH
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Binary supermassive black hole SED

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  - Inner radius of the primary SMBH
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- Parameters for the binary BH model

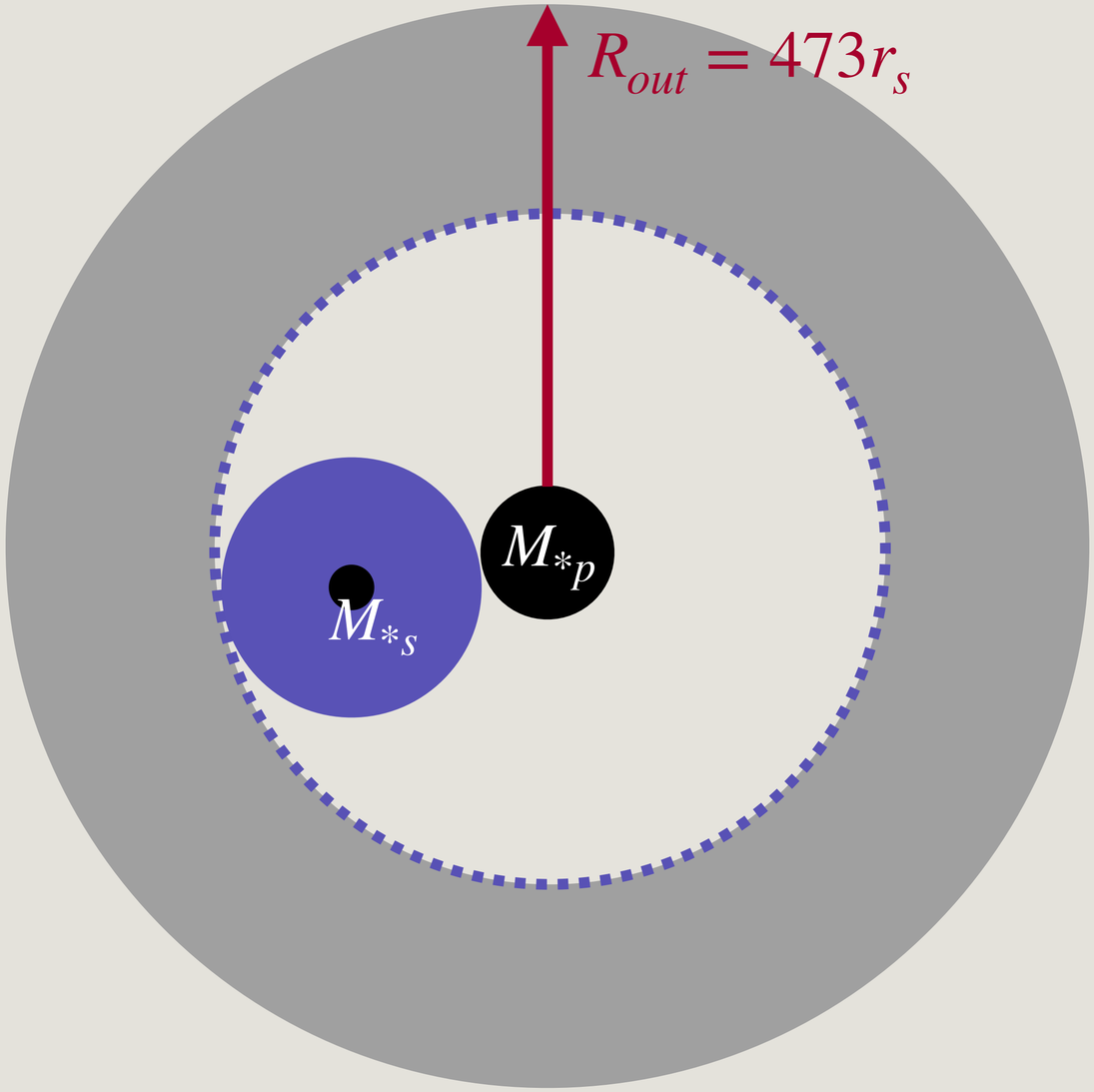
Primary SMBH

Roche lobe fraction

$$\boldsymbol{\theta} \equiv \{M_{*,p}, \alpha_p, p_p, \theta_p \; ; \; M_{*,s}, \alpha_s, p_s, \theta_s \; ; \; f_{r,s}, a_{BBH}\}$$

Secondary SMBH

Semi major axis



II . Modele Spectral Energy Distribution - Accretion disk

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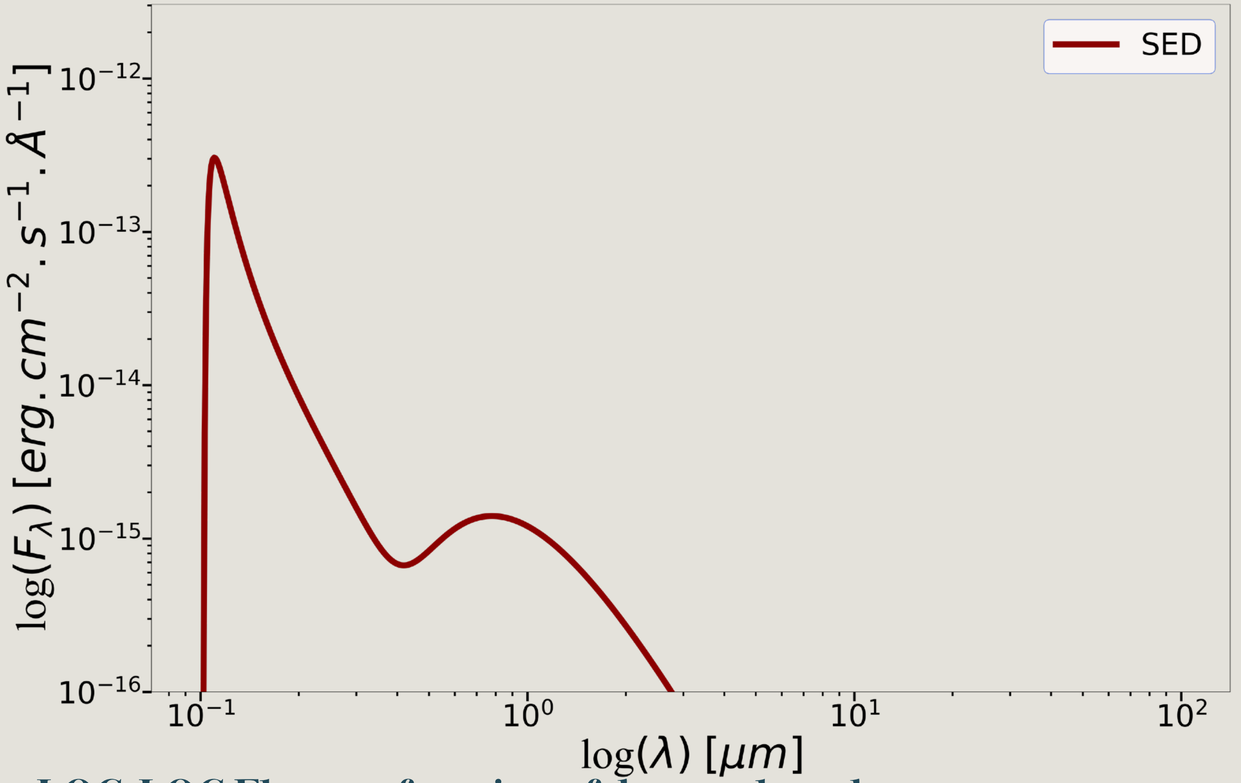
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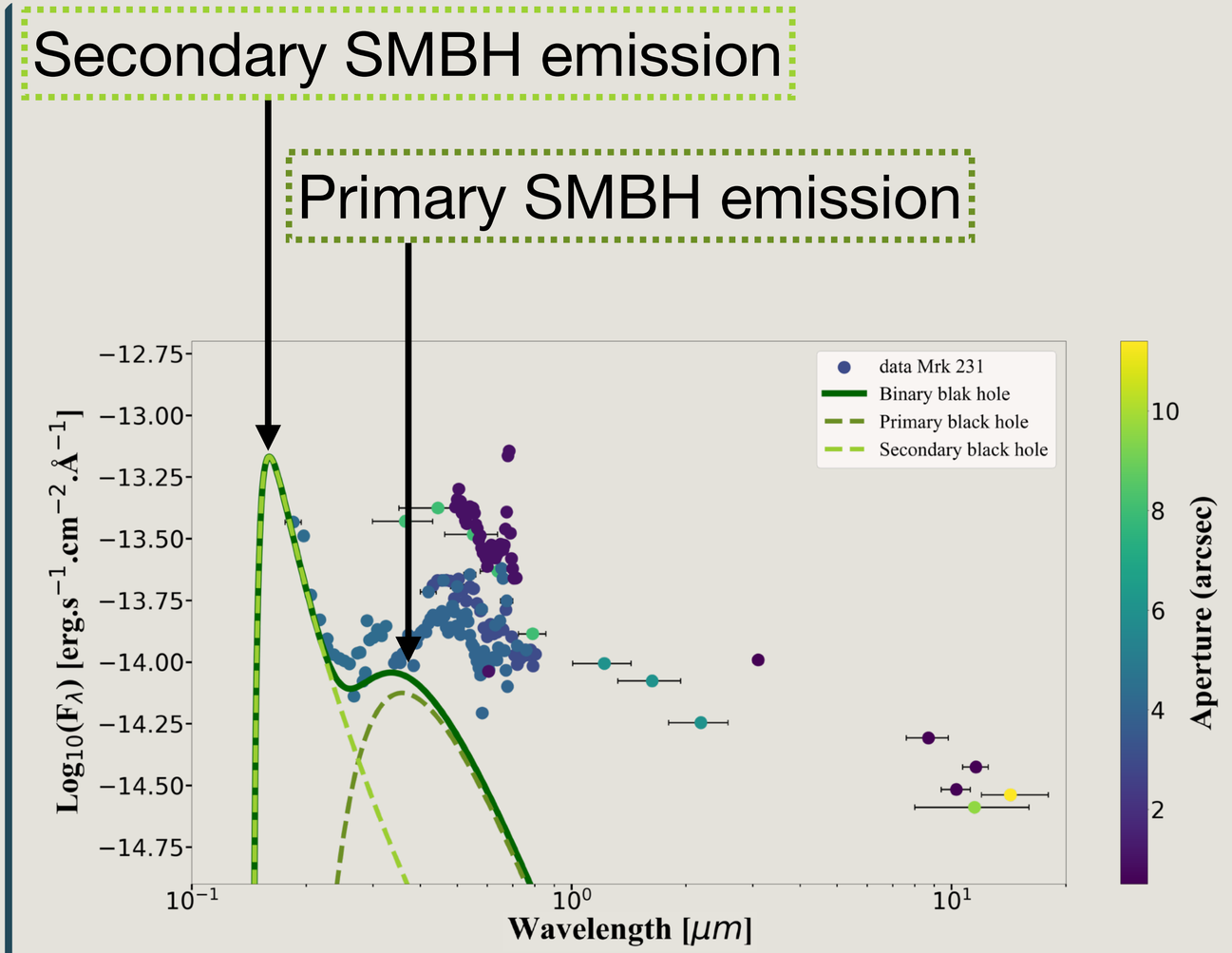


LOG-LOG Flux as a function of the wavelength. Binary SMBH

Biedermann et al. (submitted)

II . Modele Spectral Energy Distribution - Component insertion

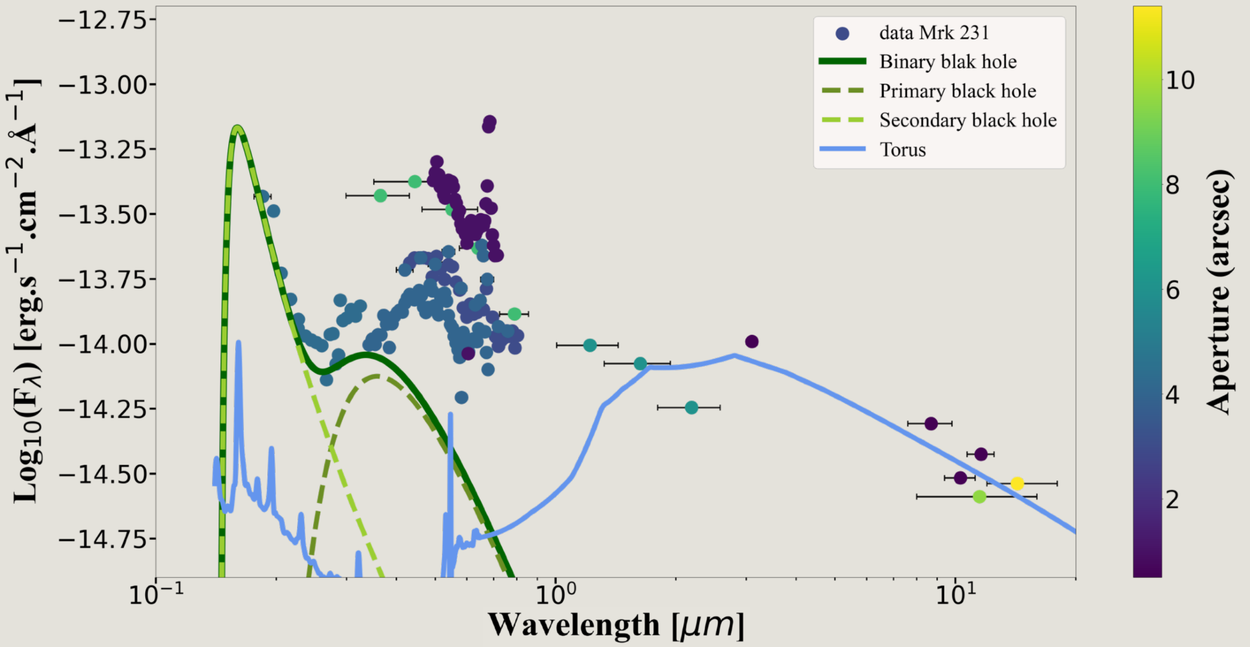
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LOG-LOG Flux as a function of the wavelength

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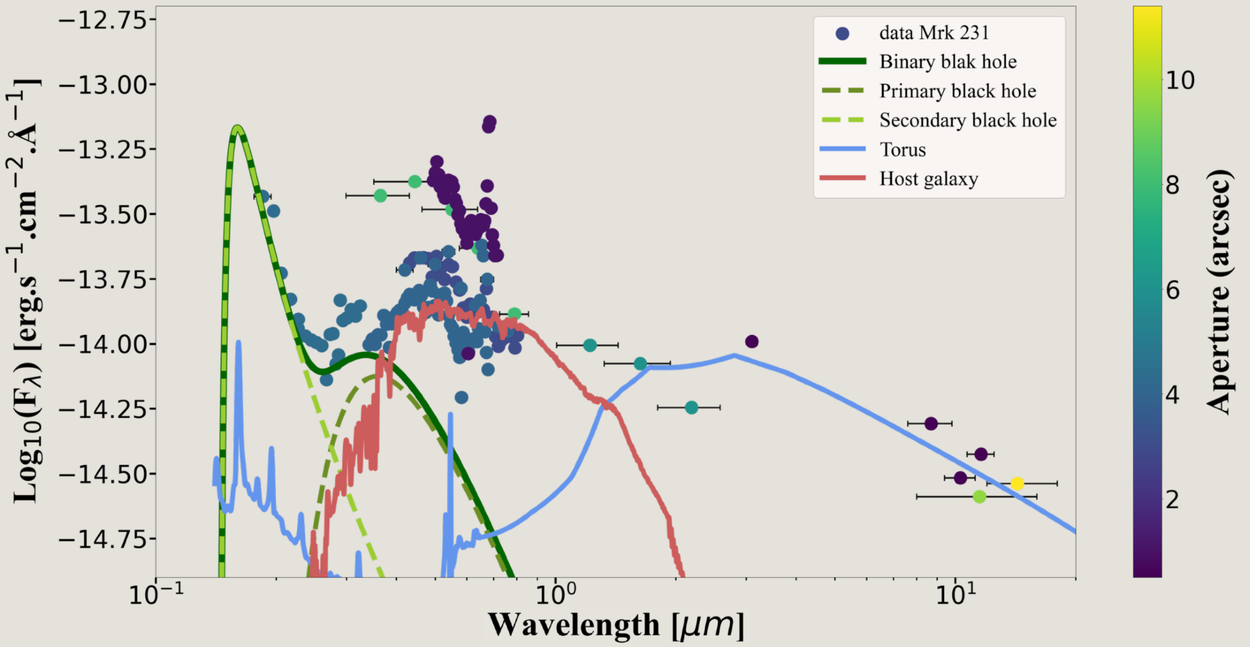
- Binary SMBH
- Torus *(Siebenmorgen et al. 2016)*



LOG-LOG Flux as a function of the wavelength

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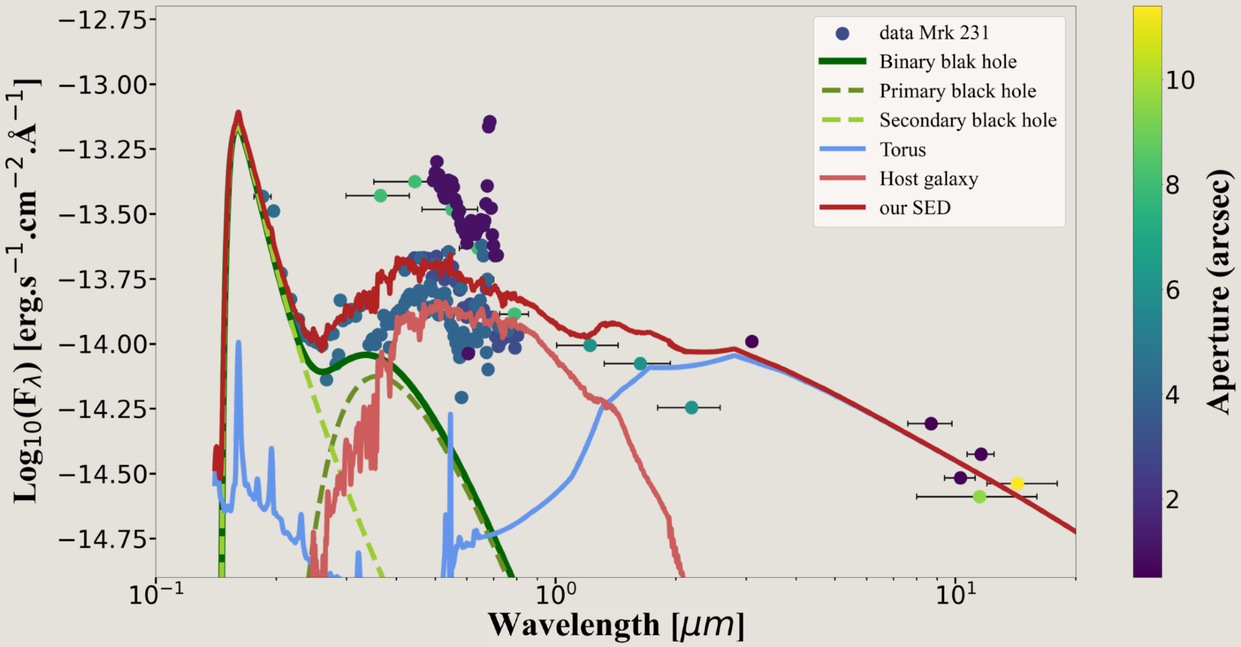
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- Host galaxy *(Polletta et al. 2007)*



LOG-LOG Flux as a function of the wavelength

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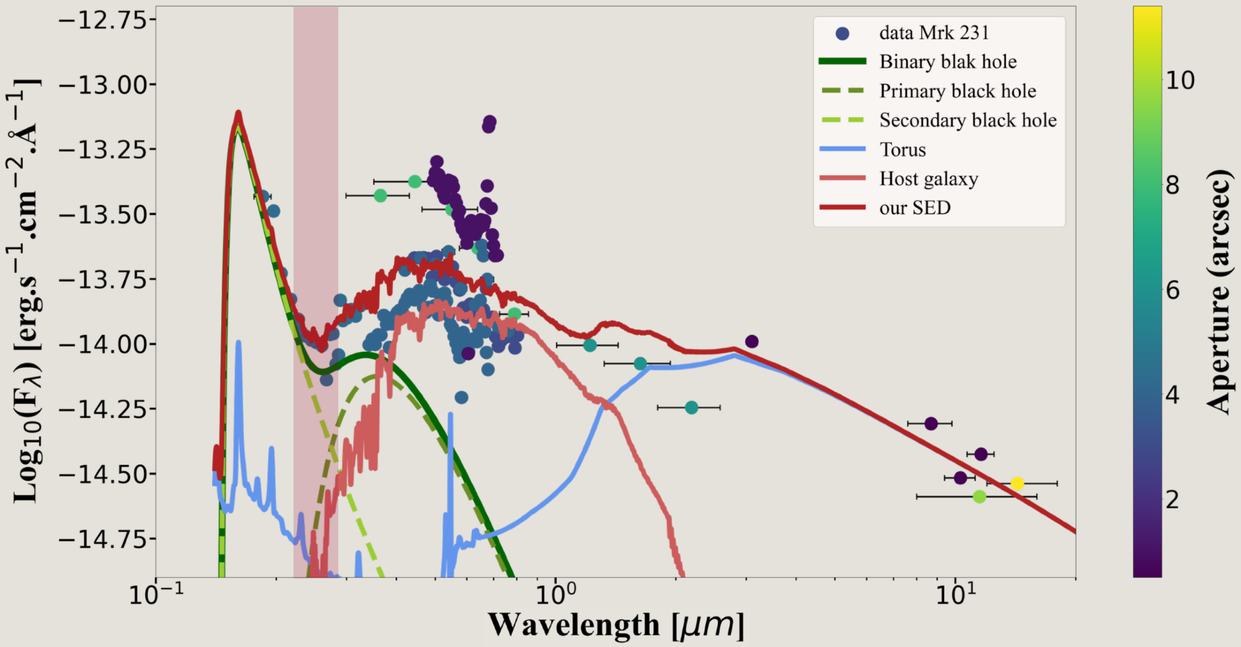
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LOG-LOG Flux as a function of the wavelength
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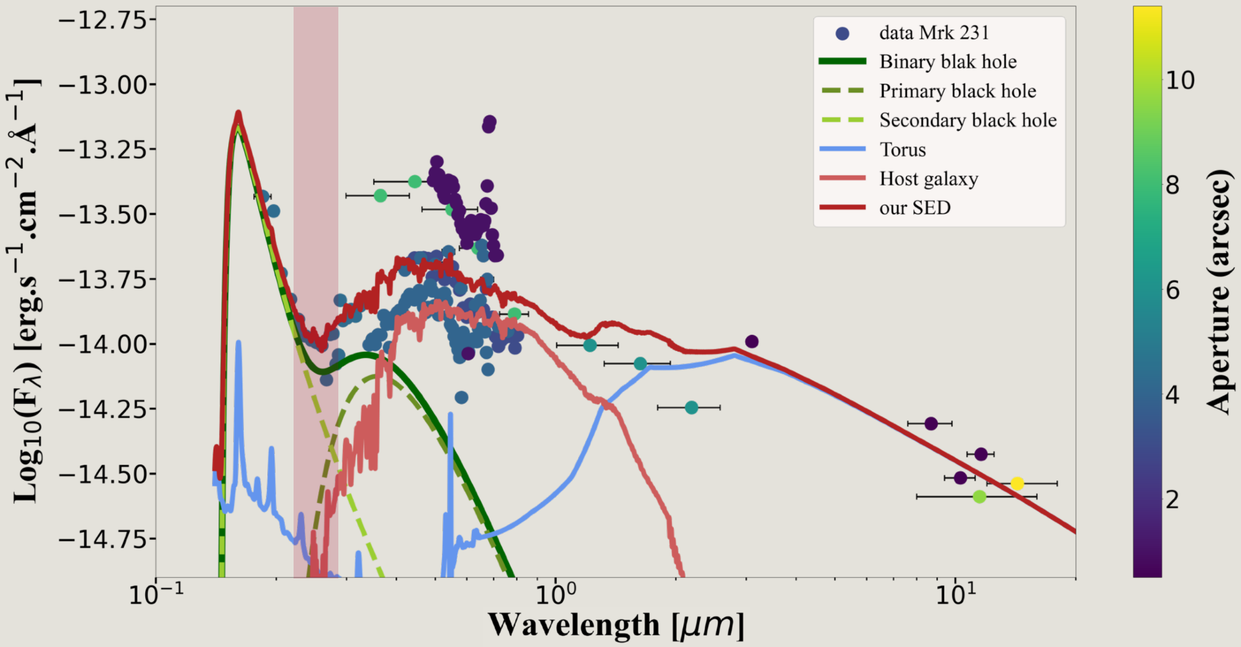


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➡ **A step further : what is the response of the model in polarization ?**



LOG-LOG Flux as a function of the wavelength
Biedermann et al. (submitted)

Polarimetric analysis

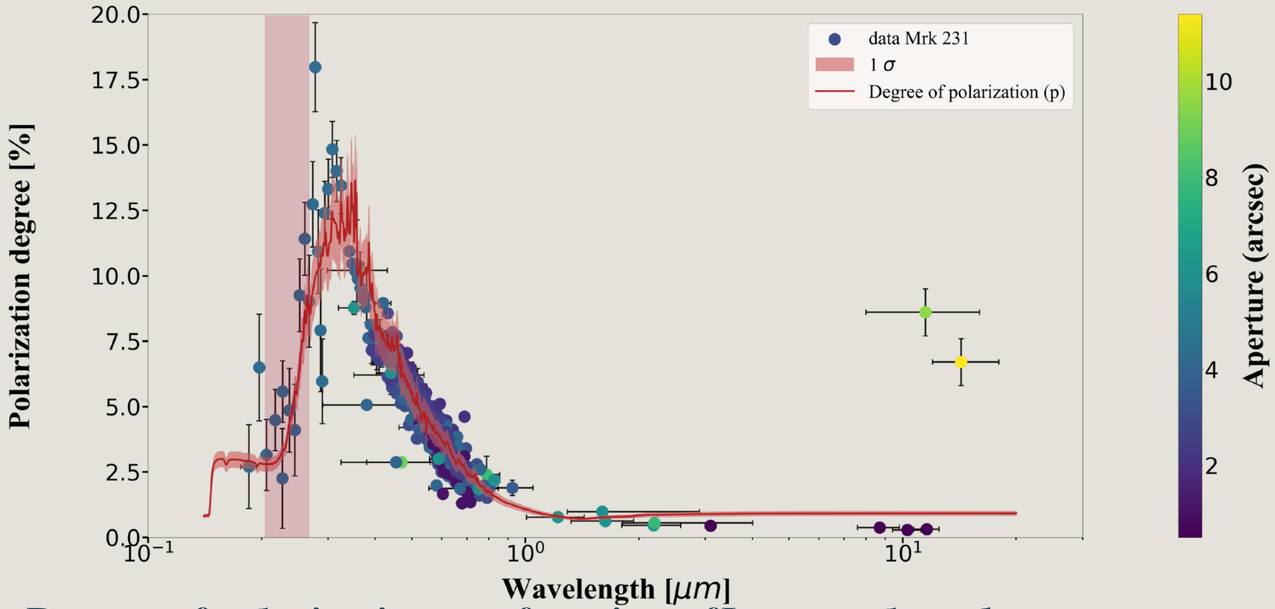
- For each component of the SED, a best fit polarization value is assigned and the Stokes fluxes are summed

III . Polarimetry - Degree of polarization

Polarimetric analysis

- For each component of the SED, a best fit polarization value is assigned and the Stokes fluxes are summed

- Polarization degree: $$p = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}$$



Degree of polarization as a function of Log wavelength
Biedermann et al. (submitted)

	Primary Black hole	Secondary Black hole	Host galaxy	Torus
P (%)	23.0 ± 2.7	3.1 ± 0.3	0	0.9 ± 0.1

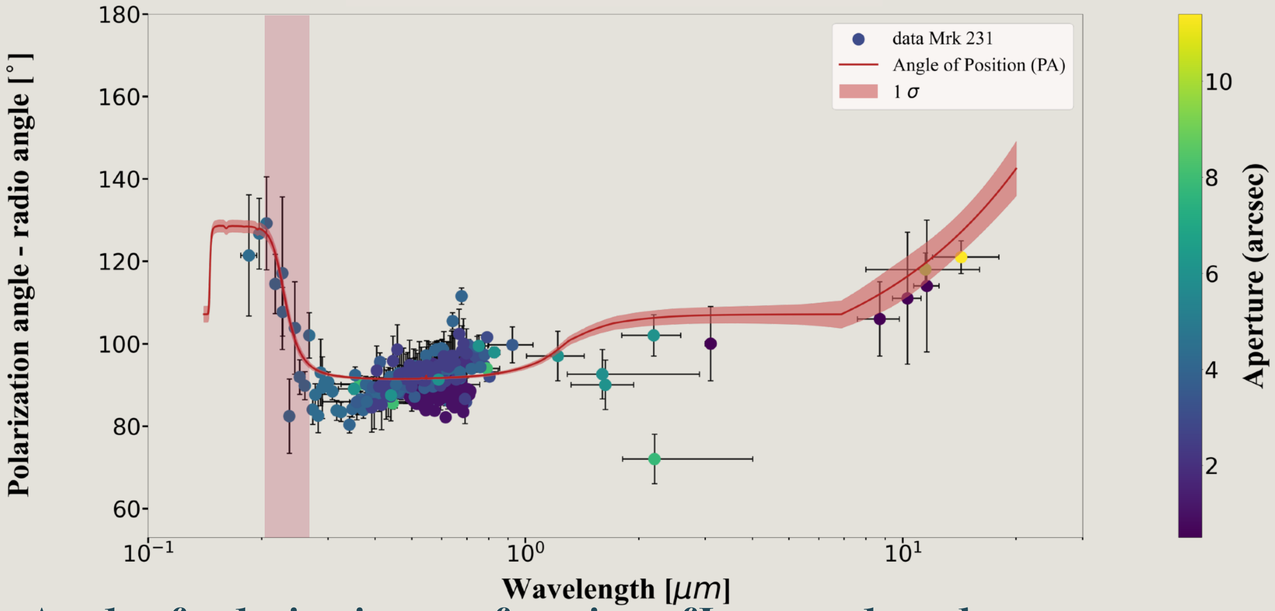
III . Polarimetry - Angle of polarization

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Angle of polarization as a function of Log wavelength
Biedermann et al. (submitted)

	Primary Black hole	Secondary Black hole	Host galaxy	Torus
P (%)	23.0 ± 2.7	3.1 ± 0.3	0	0.9 ± 0.1
PA (°)	95.9 ± 0.2	133.8 ± 1.5	0	112.1 ± 2.1

III . Polarimetry - Polarized flux

Polarimetric analysis

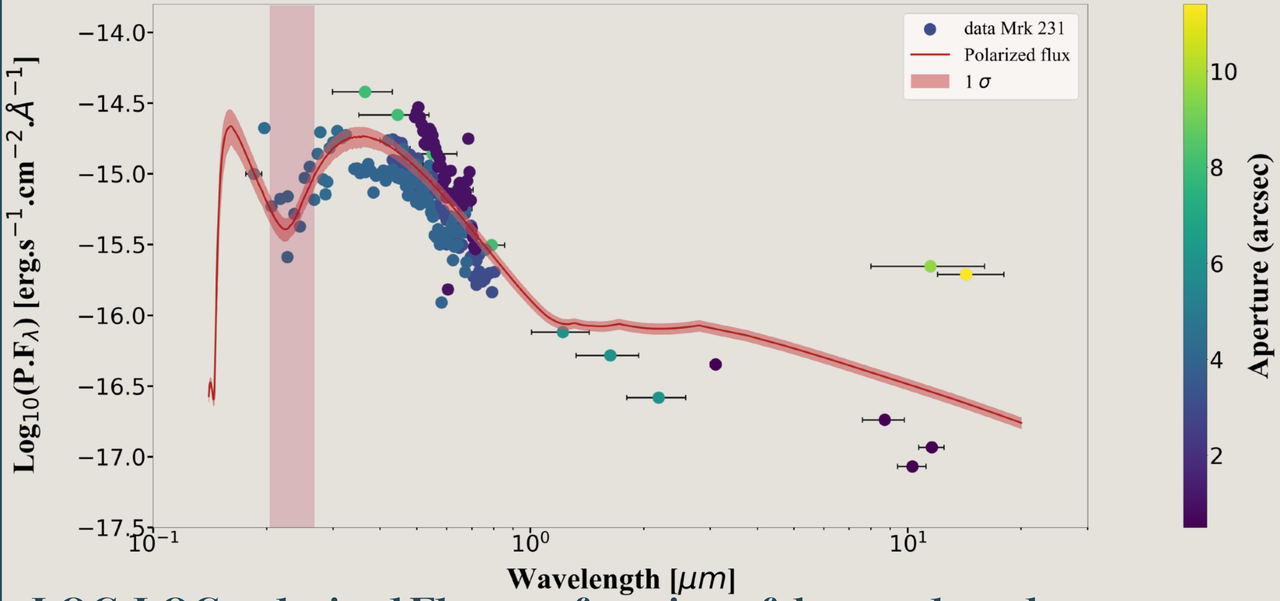
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- Polarized flux

  Strong prediction for a second bump in FUV



LOG-LOG polarized Flux as a function of the wavelength
Biedermann et al. (submitted)

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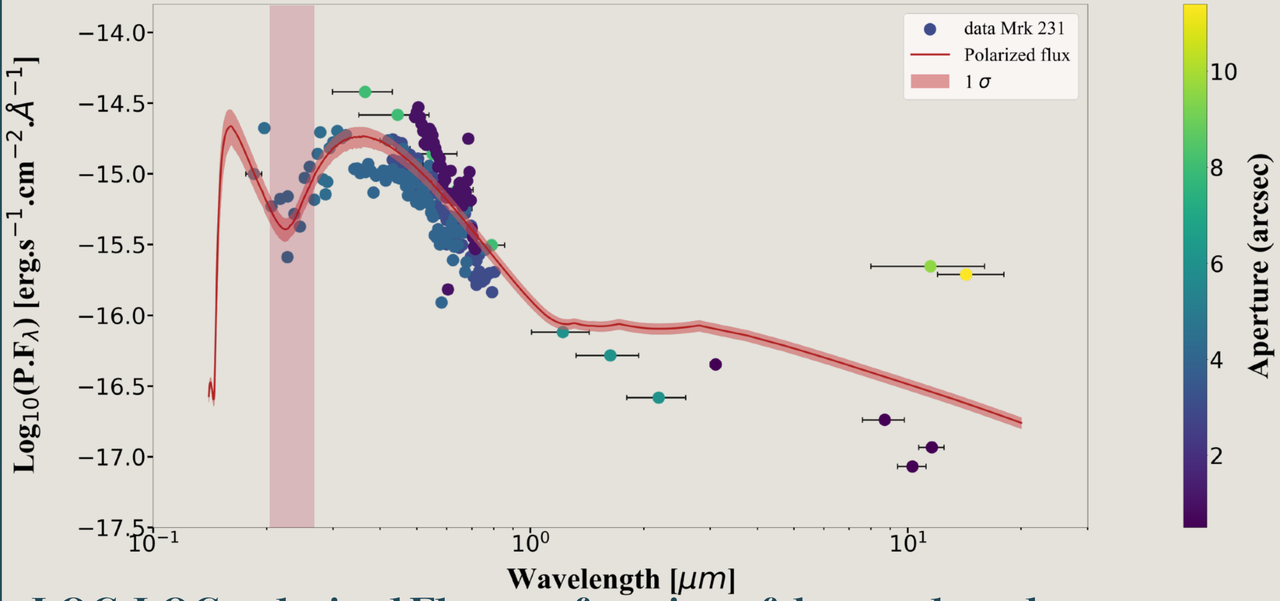
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Model reproduces three independent measurements



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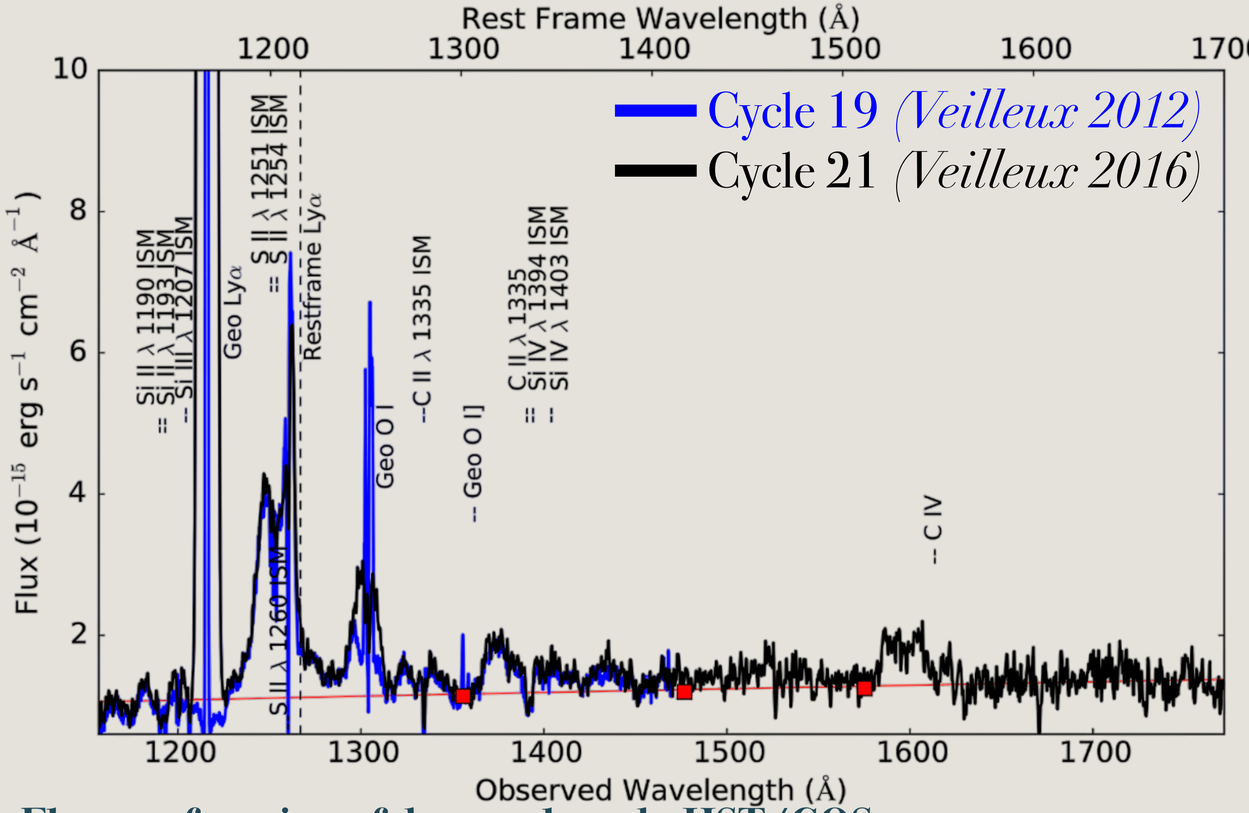
CONCLUSION

➡ Binary model of SMBH in agreement with our spectropolarimetric data

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- Arguments in tension with the binary hypothesis
  - Stability in time for the continuum emission *(Veilleux et al. 2016)*



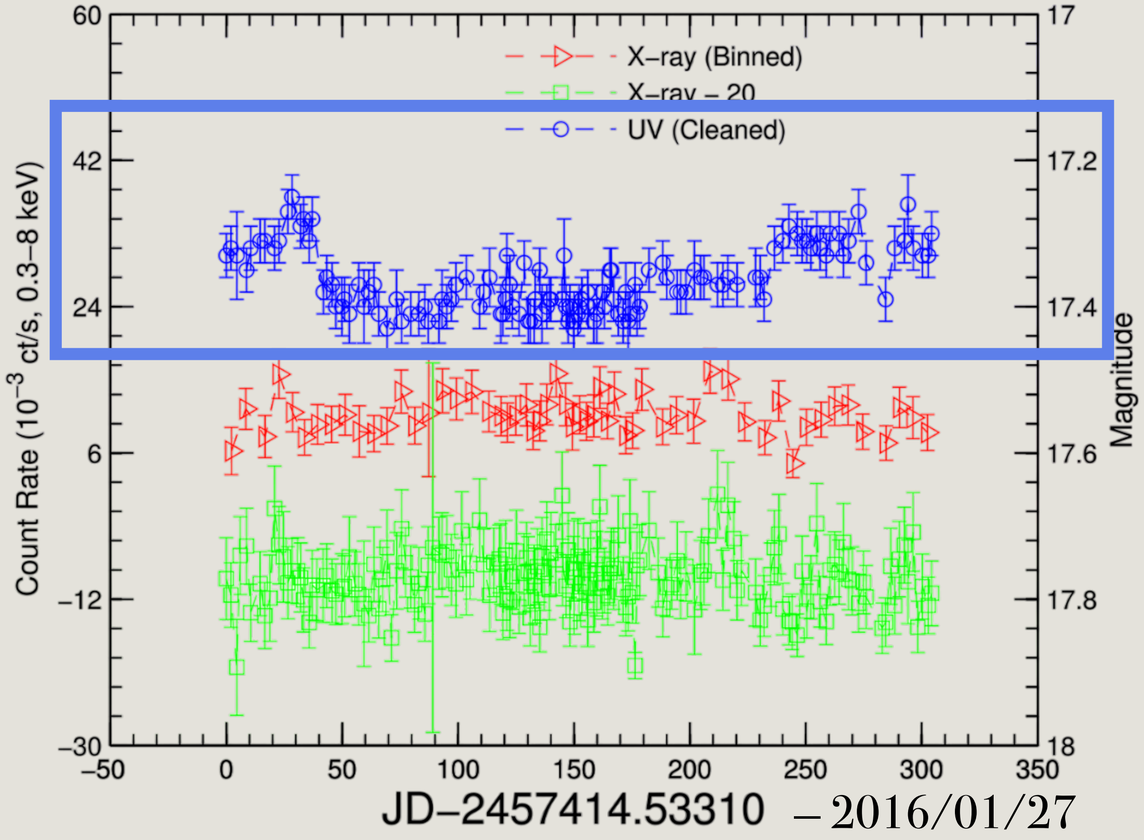
Flux as a function of the wavelength, HST/COS
Veilleux et al. (2016)

CONCLUSION

➡ Binary model of SMBH in agreement with our spectropolarimetric data

➡ Arguments in tension with the binary hypothesis

- ❖ Stability in time for the continuum emission *(Veilleux et al. 2016)*
- ❖ Variability of the UV continuum emission recently discovered *(Wang et al. 2018)*



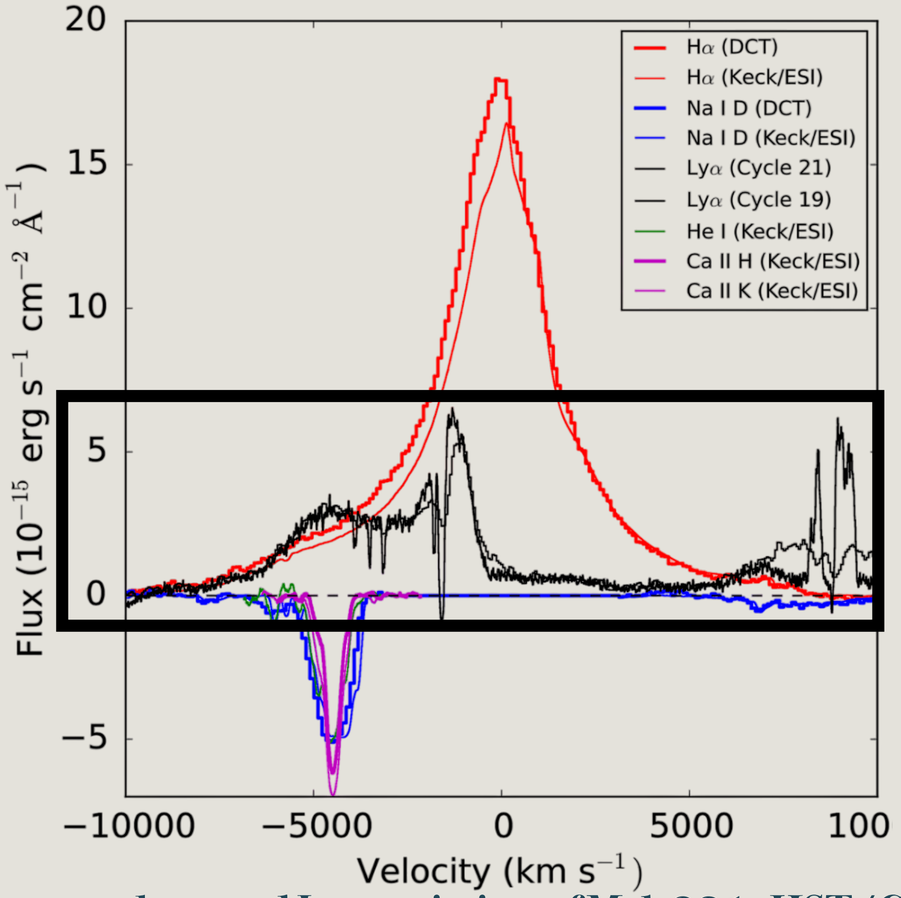
Light curves of Mrk 231 in uvm2 filter centered at 2246 Å , Swift
Wang et al. (2018)

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➡ Arguments in tension with the binary hypothesis

- ❖ Stability in time for the continuum emission *(Veilleux et al. 2016)*
- ❖ Variability of the UV continuum emission recently discovered *(Wang et al. 2018)*
- ❖ Stable Lyα profile *(Veilleux et al. 2016)*



Continuum-subtracted Lyα emission of Mrk 231, HST/COS
Veilleux et al. (2016)

CONCLUSION

➡ Binary model of SMBH in agreement with our spectropolarimetric data

➡ Arguments in tension with the binary hypothesis

➡ Let's build a POLLUX-like instrument

✓ To probe the variation in time of Lyα profile ($R \geq 20000$; $\lambda_{rest} \sim 1170 - 1250$ Å)

✓ To confirm/reject the presence of a second bump in FUV most easily detectable in polarized light ($\lambda_{rest} \sim 1460 - 2350$ Å)

✓ And use POLLUX as powerful detector of binaries SMBH in AGNs

Thank you

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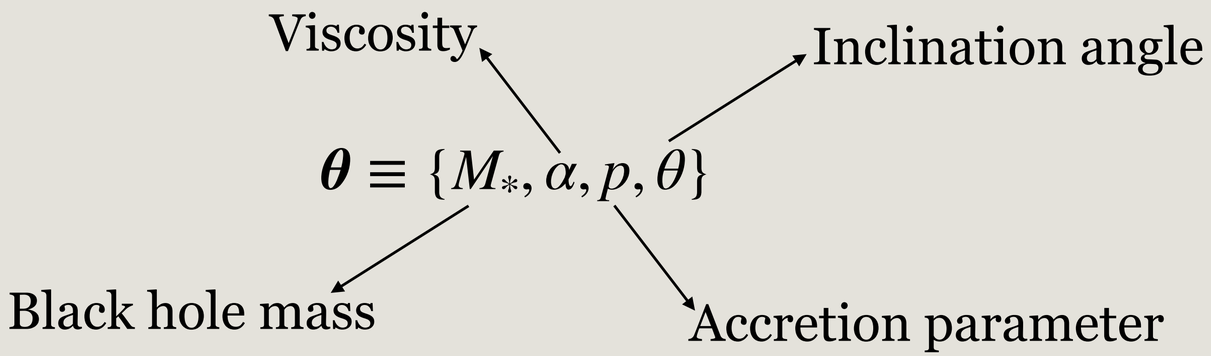
- Theoretical radius - Toomre parameter indicates a local gravitational instability in the disc
  - Disk become unstable : $\dot{M} > \dot{M}_{+}$ *Lacosta (2016)*

$$\dot{M}_{+} = 8.07 \times 10^{15} \alpha_{0.1}^{-0.01} \left(\frac{r}{10^{10}[cm]} \right)^{2.64} \left(\frac{M_*}{M_\odot} \right)^{-0.89}$$

$$\dot{M} = \alpha p \dot{M}_{Eddington}$$

- Black hole masse : M_*
- Viscosity : α
- Accretion parameter : p
- Radius of the accretion disk : r
- Eddington accretion rate : $\dot{M}_{Eddington}$

- $\dot{M}$ is linked to thermal instabilities



BACK SLIDE

❖Inner radius for the primary SMBH

$$R_{int,p} = \frac{a_{BBH}}{1+q} + a_{BBH}\left(\frac{M_{*,s}}{3M_{*,p}}\right)^{\frac{1}{3}}$$

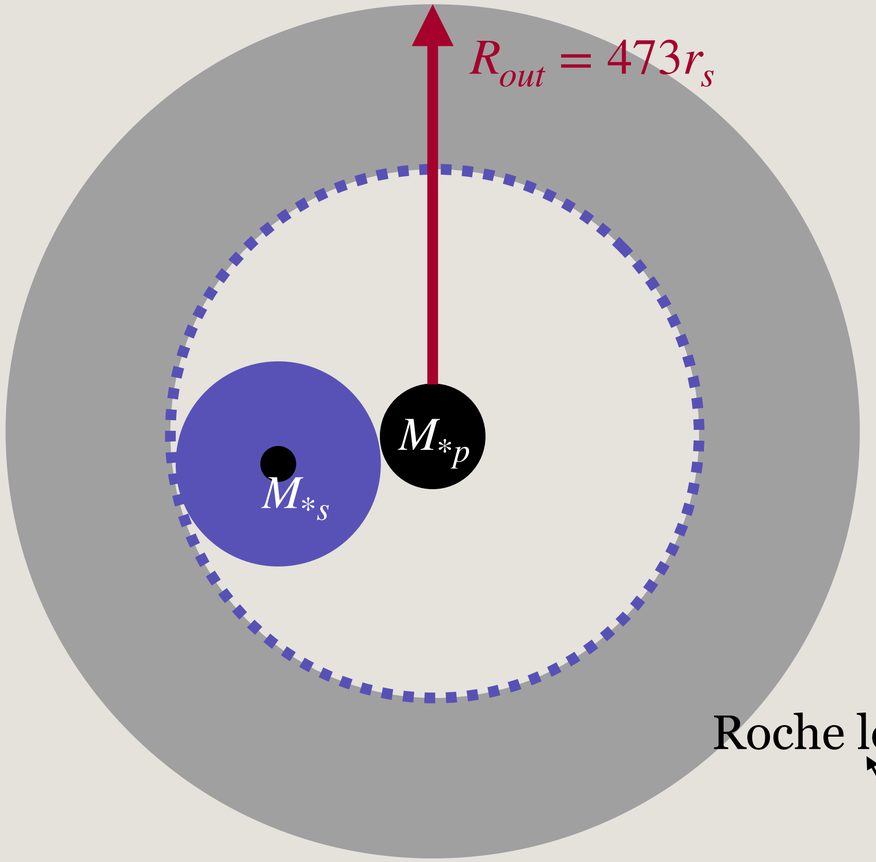
Yan et al (2015)

❖Outer radius for the secondary SMBH

$$R_{out,s} = f_{r,s} \times \left(\frac{0.49 a_{BBH} q^{\frac{2}{3}}}{0.6 q^{\frac{2}{3}} + log(1+q^{\frac{1}{2}})}\right)$$

Yan et al (2015)

$$R_{int,s} = r_s$$



Roche lobe fraction

$$\boldsymbol{\theta} \equiv \{\underbrace{M_{*,p}, \alpha_p, p_p, \theta_p}_{\text{Primary SMBH}} ; \underbrace{M_{*,s}, \alpha_s, p_s, \theta_s}_{\text{Secondary SMBH}} ; f_{r,s}, a_{BBH}\}$$

Semi major axis

II . Modele Spectral Energy Distribution - Accretion disk

Binary supermassive black hole SED

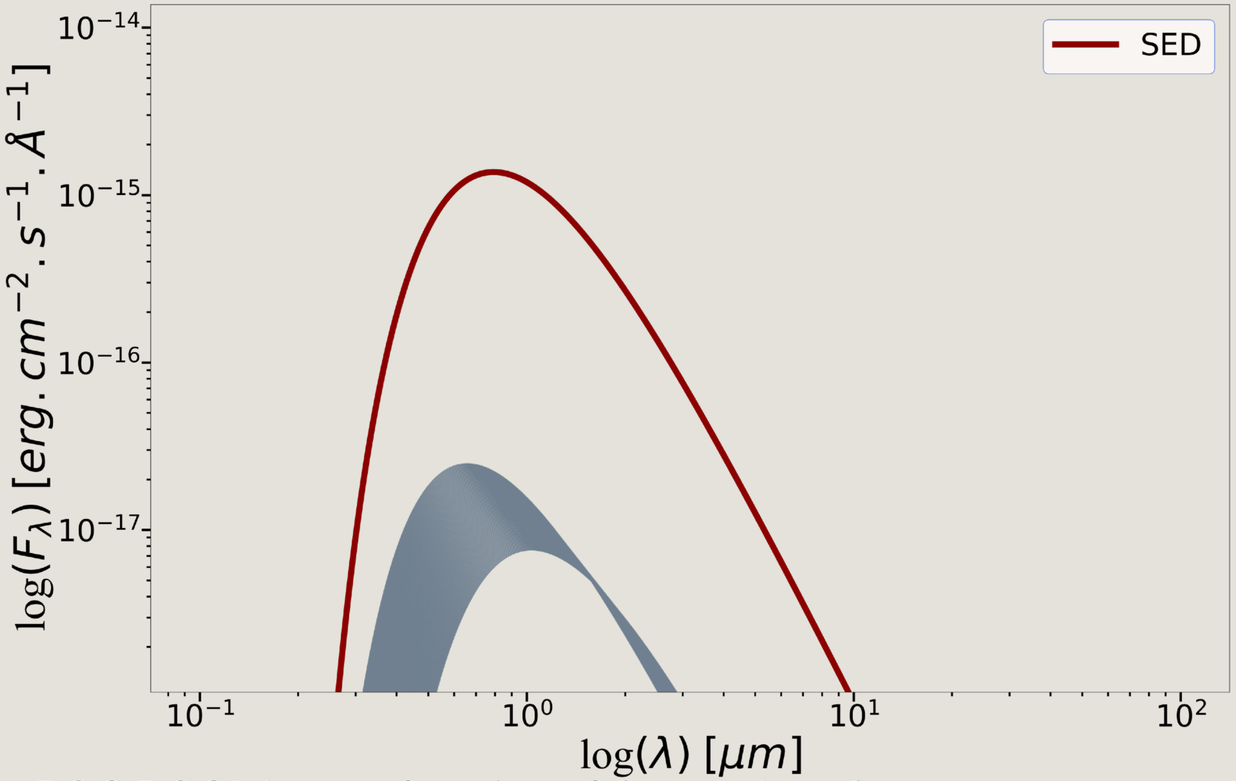
- Add secondary SMBH
  - Inner radius for the primary SMBH
  - Outer radius for the secondary SMBH

- Parameter of the binary SED model

$$\boldsymbol{\theta} \equiv \{\underbrace{M_{*,p}, \alpha_p, p_p, \theta_p}_{\text{Primary SMBH}} ; \underbrace{M_{*,s}, \alpha_s, p_s, \theta_s}_{\text{Secondary SMBH}} ; f_{r,s}, a_{BBH}\}$$

Roche lobe fraction

Semi major axis



LOG-LOG Flux as a function of the wavelength
Primary SMBH $M_{*p} = 1.5 \times 10^8 M_\odot$

Biedermann et al (submitted)

II . Modele Spectral Energy Distribution - Accretion disk

Binary supermassive black hole SED

- Add secondary SMBH
  - Inner radius for the primary SMBH
  - Outer radius for the secondary SMBH

- Parameter of the binary SED model

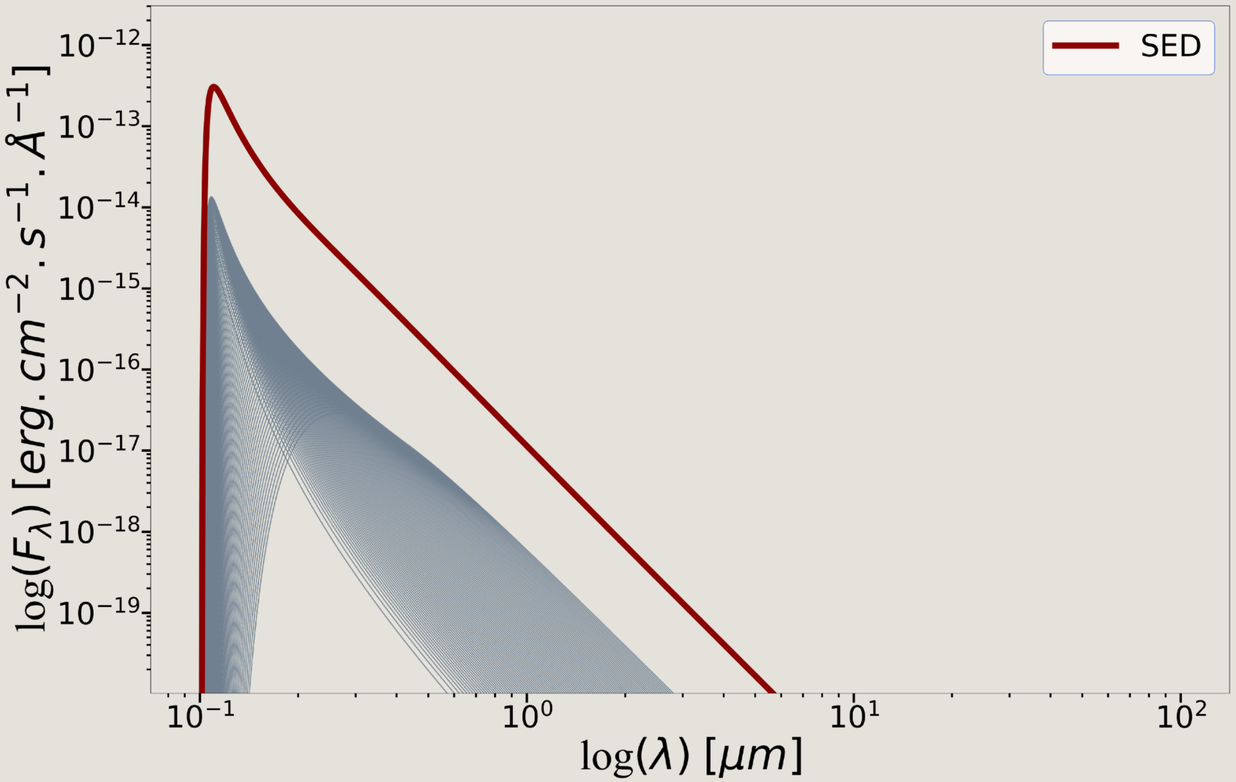
Roche lobe fraction

Primary SMBH

$$\boldsymbol{\theta} \equiv \{M_{*,p}, \alpha_p, p_p, \theta_p \; ; \; M_{*,s}, \alpha_s, p_s, \theta_s \; ; \; f_{r,s}, a_{BBH}\}$$

Secondary SMBH

Semi major axis



LOG-LOG Flux as a function of the wavelength
Secondary SMBH $M_{*s} = 4.5 \times 10^6 M_\odot$

Biedermann et al (submitted)